

LINUX COMMANDS

REFERENCE GUIDE · Common Commands & What They Do

This reference covers 286 commands across 16 categories. Commands are shown with common usage patterns. Use **man <command>** for full details.

■ FILE & DIRECTORY MANAGEMENT

ls	List files and directories in current folder
ls -la	List all files (including hidden) with details
cd <dir>	Change into a directory
cd ..	Go up one directory level
cd ~	Go to home directory
pwd	Print current working directory path
mkdir <dir>	Create a new directory
mkdir -p a/b	Create nested directories at once
rmdir <dir>	Remove an empty directory
rm <file>	Delete a file
rm -rf <dir>	Delete a directory and all its contents (careful!)
cp <src> <dst>	Copy a file from source to destination
cp -r <s> <d>	Copy a directory recursively
mv <src> <dst>	Move or rename a file/directory
touch <file>	Create an empty file or update its timestamp
ln -s <f> <l>	Create a symbolic (soft) link
find <path> -name <pat>	Search for files matching a pattern
locate <file>	Quickly find a file by name (uses a database)
tree	Display directory structure as a tree
du -sh <dir>	Show disk usage of a directory (human-readable)
stat <file>	Display detailed info about a file
file <file>	Detect and display the type of a file

■ VIEWING & READING FILES

<code>cat <file></code>	Print entire file contents to screen
<code>less <file></code>	View file page by page (press q to quit)
<code>more <file></code>	View file one screen at a time
<code>head <file></code>	Show first 10 lines of a file
<code>head -n 20</code>	Show first 20 lines of a file
<code>tail <file></code>	Show last 10 lines of a file
<code>tail -f <file></code>	Follow a file in real-time (great for logs)
<code>tail -n 50</code>	Show last 50 lines of a file
<code>wc <file></code>	Count lines, words, and characters in a file
<code>wc -l <file></code>	Count only lines in a file
<code>diff <f1> <f2></code>	Compare two files line by line
<code>xxd <file></code>	Show file in hex dump format

■ TEXT PROCESSING

<code>grep <pat> <f></code>	Search for a pattern inside a file
<code>grep -r <pat></code>	Search recursively through directories
<code>grep -i <pat></code>	Case-insensitive search
<code>grep -n <pat></code>	Show line numbers with matches
<code>grep -v <pat></code>	Show lines that do NOT match the pattern
<code>sed 's/a/b/g'</code>	Replace all occurrences of 'a' with 'b' in a file
<code>awk '{print \$1}'</code>	Print the first column/field from each line
<code>cut -d: -f1</code>	Cut fields from text using a delimiter
<code>sort <file></code>	Sort lines of a file alphabetically
<code>sort -n <file></code>	Sort lines numerically
<code>sort -r <file></code>	Sort lines in reverse order
<code>sort -u <file></code>	Sort and remove duplicate lines
<code>uniq <file></code>	Remove consecutive duplicate lines
<code>uniq -c</code>	Count how many times each line repeats
<code>tr 'a-z' 'A-Z'</code>	Translate characters (e.g., lowercase to uppercase)
<code>tee <file></code>	Read from stdin and write to both file and stdout
<code>xargs</code>	Build and run commands from standard input
<code>paste <f1> <f2></code>	Merge lines from two files side by side
<code>join <f1> <f2></code>	Join lines of two files on a common field
<code>comm <f1> <f2></code>	Compare two sorted files line by line

■ FILE PERMISSIONS & OWNERSHIP

chmod 755 <f>	Set file permissions using numeric (octal) mode
chmod +x <f>	Make a file executable
chmod -w <f>	Remove write permission from a file
chown user <f>	Change the owner of a file
chown u:g <f>	Change owner and group of a file
chgrp <grp> <f>	Change the group ownership of a file
umask	Display or set default permission mask for new files
ls -l	View permissions, owner, and group of files
getfacl <file>	View Access Control List (ACL) of a file
setfacl	Set ACL permissions on a file or directory

■ PROCESS MANAGEMENT

ps	Show currently running processes for this session
ps aux	Show all running processes for all users
ps aux grep x	Find a specific running process by name
top	Live interactive view of running processes and CPU/RAM
htop	Enhanced interactive process viewer (like top, better UI)
kill <PID>	Send a termination signal to a process by its PID
kill -9 <PID>	Force kill a process immediately (SIGKILL)
killall <name>	Kill all processes with a given name
pkill <name>	Kill processes by name pattern
jobs	List background and suspended jobs in current shell
bg	Resume a suspended job in the background
fg	Bring a background job to the foreground
nohup <cmd> &	Run a command immune to hangup, in the background
nice -n 10 cmd	Run a command with lower CPU priority
renice 10 <PID>	Change priority of a running process
pgrep <name>	Find the PID of a process by name
pidof <name>	Get the PID(s) of a running program
strace <cmd>	Trace system calls made by a process
lsof	List all open files and the processes using them
lsof -i :80	Show what process is using port 80
wait	Wait for background processes to finish

■ NETWORKING

ping <host>	Test connectivity to a host by sending ICMP packets
ping -c 4 host	Send exactly 4 ping packets
tracert <h>	Trace the path packets take to reach a host
tracert <h>	Trace path to host (no root needed)
ip a	Show all network interfaces and their IP addresses
ip link	Show or manage network interfaces
ip route	Show the routing table
ifconfig	Display or configure network interfaces (older tool)
netstat -tuln	Show all listening ports and their protocols
ss -tuln	Modern replacement for netstat — show open sockets
curl <url>	Transfer data from or to a URL (HTTP, FTP, etc.)
curl -O <url>	Download a file from a URL
wget <url>	Download files from the internet non-interactively
wget -r <url>	Recursively download a website
ssh user@host	Connect securely to a remote machine via SSH
ssh-keygen	Generate SSH public/private key pair
ssh-copy-id	Copy SSH public key to a remote server
scp f user@h:	Securely copy a file to a remote host
rsync -av s d	Sync files/directories efficiently over network or locally
nmap <host>	Scan a host for open ports and services
nmap -sV host	Detect service versions on open ports
dig <domain>	Query DNS for a domain name (detailed)
nslookup <dom>	Query DNS for a domain name (simpler)
host <domain>	Look up DNS info for a domain
whois <domain>	Look up registration info for a domain
nc -zv host port	Check if a port is open on a host (netcat)
iptables -L	List current firewall rules
ufw status	Check UFW firewall status (Ubuntu)
ufw allow 22	Allow SSH through UFW firewall
hostname	Display or set the system hostname
hostname -I	Show the machine's IP address(es)
arp -n	Display the ARP table (IP to MAC mapping)
route -n	Show the kernel routing table (older command)
tcpdump -i eth0	Capture network packets on an interface

iwconfig	Show wireless network interface configuration
nmcli	NetworkManager CLI to manage network connections

■ DISK & STORAGE

df -h	Show disk space usage for all mounted filesystems
du -sh <dir>	Show total size of a directory
du -ah <dir>	Show size of every file in a directory
lsblk	List all block devices (disks, partitions, USBs)
fdisk -l	List all disks and their partition tables
mount	Show all currently mounted filesystems
mount /dev/sdb1 /mnt	Mount a disk partition to a directory
umount /mnt	Unmount a mounted filesystem
mkfs.ext4 /dev/sdb1	Format a partition with ext4 filesystem
fsck /dev/sdb1	Check and repair a filesystem
blkid	Show UUIDs and filesystem types of block devices
parted	Partition management tool (interactive or scripted)
dd if=/dev/sda of=backup.img	Clone a disk or create a disk image
sync	Flush filesystem buffers to disk
iostat	Show CPU and disk I/O statistics
hdparm -I /dev/sda	Display detailed info about a hard disk

■ ARCHIVING & COMPRESSION

tar -cvf a.tar dir	Create a tar archive of a directory
tar -xvf a.tar	Extract a tar archive
tar -czvf a.tar.gz dir	Create a compressed .tar.gz archive
tar -xzvf a.tar.gz	Extract a .tar.gz archive
tar -cjvf a.tar.bz2 dir	Create a .tar.bz2 (bzip2) archive
tar -xjvf a.tar.bz2	Extract a .tar.bz2 archive
gzip <file>	Compress a file with gzip (replaces original)
gunzip <file>	Decompress a .gz file
bzip2 <file>	Compress a file with bzip2
bunzip2 <file>	Decompress a .bz2 file
xz <file>	Compress with xz (better compression ratio)
unxz <file>	Decompress a .xz file
zip -r a.zip dir	Create a zip archive of a directory
unzip a.zip	Extract a zip archive
unzip -l a.zip	List contents of a zip without extracting
7z a arc.7z dir	Create a 7zip archive (high compression)
7z x arc.7z	Extract a 7zip archive

■ USER & GROUP MANAGEMENT

whoami	Print the current logged-in username
who	Show who is currently logged in to the system
w	Show logged-in users and what they are doing
id	Display user ID, group ID, and group memberships
useradd <user>	Create a new user account
useradd -m <u>	Create a new user with a home directory
userdel <user>	Delete a user account
usermod -aG grp user	Add a user to a group
passwd <user>	Change the password of a user
groupadd <grp>	Create a new group
groupdel <grp>	Delete a group
groups <user>	Show which groups a user belongs to
su <user>	Switch to another user account
sudo <cmd>	Run a command as superuser (root)
sudo -i	Open an interactive root shell
visudo	Safely edit the sudoers file
last	Show a history of user logins
lastlog	Show the most recent login for each user
finger <user>	Display info about a user
chage -l <user>	Show password expiry info for a user

■ SYSTEM INFORMATION

uname -a	Show all system information (kernel, arch, hostname)
uname -r	Show only the kernel version
hostname	Display the system hostname
uptime	Show how long the system has been running
date	Display the current date and time
cal	Display a calendar for the current month
timedatectl	Show or set the system date, time, and timezone
free -h	Show RAM and swap memory usage (human-readable)
vmstat	Show memory, CPU, swap, and I/O statistics
lscpu	Show detailed CPU architecture information
lshw	List detailed hardware information
lspci	List all PCI devices
lsusb	List all USB devices connected to the system
dmidecode	Show hardware info from BIOS/UEFI (DMI table)
cat /proc/cpuinfo	Show detailed CPU info from kernel
cat /proc/meminfo	Show detailed memory info from kernel
dmesg	Show kernel ring buffer messages (boot logs, hardware)
dmesg tail	Show the most recent kernel messages
journalctl	Query and view systemd journal logs
journalctl -xe	Show recent journal entries with explanations
env	Display all current environment variables
printenv PATH	Print a specific environment variable
arch	Show the machine architecture (e.g., x86_64)
nproc	Show the number of CPU cores available

■ PACKAGE MANAGEMENT (Debian/Ubuntu)

apt update	Refresh the list of available packages
apt upgrade	Upgrade all installed packages to latest version
apt install x	Install a package
apt remove x	Remove a package (keep config files)
apt purge x	Remove a package and all its config files
apt autoremove	Remove unused dependency packages
apt search x	Search for a package by name or keyword
apt show x	Show detailed info about a package
apt list --installed	List all currently installed packages
dpkg -i pkg.deb	Install a .deb package file manually
dpkg -l	List all installed packages
dpkg -L <pkg>	List files installed by a package
snap install x	Install a snap package
snap list	List installed snap packages
snap remove x	Remove a snap package

■ SHELL & SCRIPTING

echo <text>	Print text or variable values to the terminal
echo \$VAR	Print the value of an environment variable
export VAR=val	Set and export an environment variable
alias ll='ls -la'	Create a shortcut for a command
unalias ll	Remove an alias
history	Show command history
history grep x	Search command history for a keyword
!!	Repeat the last command
!n	Run command number n from history
source ~/.bashrc	Reload shell config without restarting
bash <script>	Run a shell script
chmod +x script	Make a script executable
./script.sh	Run a script in the current directory
which <cmd>	Show full path of a command's executable
whereis <cmd>	Find binary, source, and man page for a command
type <cmd>	Show how a command would be interpreted
read VAR	Read input from user into a variable
test -f <file>	Test if a file exists (used in scripts)
[-d <dir>]	Test if a directory exists (used in scripts)
set -e	Exit script immediately if a command fails
set -x	Print each command before executing (debug mode)
trap 'cmd' INT	Run a command when a signal is received
cron / crontab	Schedule commands to run automatically
crontab -e	Edit the cron schedule for the current user
crontab -l	List all scheduled cron jobs
at <time>	Schedule a one-time command at a specific time

■ REDIRECTION & PIPES

<code>cmd > file</code>	Redirect stdout output to a file (overwrite)
<code>cmd >> file</code>	Redirect stdout output to a file (append)
<code>cmd 2> file</code>	Redirect stderr (errors) to a file
<code>cmd &> file</code>	Redirect both stdout and stderr to a file
<code>cmd < file</code>	Use a file as stdin input for a command
<code>cmd1 cmd2</code>	Pipe stdout of cmd1 as stdin to cmd2
<code>cmd1 tee f</code>	Pipe output and also save it to a file
<code>cmd > /dev/null</code>	Discard output (send to null device)
<code>cmd 2>&1</code>	Redirect stderr to the same place as stdout
<code><<EOF ... EOF</code>	Heredoc: pass a block of text as stdin

■ SEARCHING & FINDING

<code>find / -name f</code>	Search entire filesystem for a filename
<code>find . -type f</code>	Find only regular files in current directory
<code>find . -type d</code>	Find only directories
<code>find . -mtime -1</code>	Find files modified in the last 1 day
<code>find . -size +1M</code>	Find files larger than 1 MB
<code>find . -perm 777</code>	Find files with specific permissions
<code>find . -exec cmd {} ;</code>	Run a command on each file found
<code>locate <file></code>	Quickly search for files using a prebuilt index
<code>updatedb</code>	Update the locate command's file database
<code>which <cmd></code>	Find the path of an executable command
<code>grep -r <pat> .</code>	Recursively search for a pattern in all files
<code>grep -l <pat></code>	Show only filenames that contain the pattern
<code>grep -c <pat></code>	Count lines that match the pattern in a file

■ SYSTEM CONTROL & SERVICES

<code>systemctl start x</code>	Start a service
<code>systemctl stop x</code>	Stop a service
<code>systemctl restart x</code>	Restart a service
<code>systemctl status x</code>	Check the status of a service
<code>systemctl enable x</code>	Enable a service to start at boot
<code>systemctl disable x</code>	Disable a service from starting at boot
<code>systemctl list-units</code>	List all active systemd units
<code>service x start</code>	Start a service (older SysV style)
<code>shutdown -h now</code>	Shut down the system immediately
<code>shutdown -r now</code>	Reboot the system immediately
<code>reboot</code>	Reboot the system
<code>halt</code>	Stop all processes and halt the system
<code>init 0</code>	Shutdown (runlevel 0)
<code>init 6</code>	Reboot (runlevel 6)
<code>runlevel</code>	Show the current and previous system runlevel
<code>wall <msg></code>	Broadcast a message to all logged-in users

■ HELP & DOCUMENTATION

<code>man <cmd></code>	Open the full manual page for a command
<code>man -k <topic></code>	Search manual pages by keyword
<code>info <cmd></code>	Read info documentation for a command
<code><cmd> --help</code>	Show a quick help summary for a command
<code>apropos <kw></code>	Find commands related to a keyword (like <code>man -k</code>)
<code>whatis <cmd></code>	Show a one-line description of a command
<code>tldr <cmd></code>	Show practical examples for a command (if installed)
<code>help <builtin></code>	Get help for a built-in shell command

*TIP: Use **Tab** for auto-completion, **Ctrl+R** to search history, **Ctrl+C** to cancel a running command, and **man <command>** to read the full manual for any command.*